Lab 04: Simple Python Database Connections

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Registration Number: 2023-BSCS-32

Course Code: CS-649

Section: A

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Task 1: Fork and Clone the Databases Repository

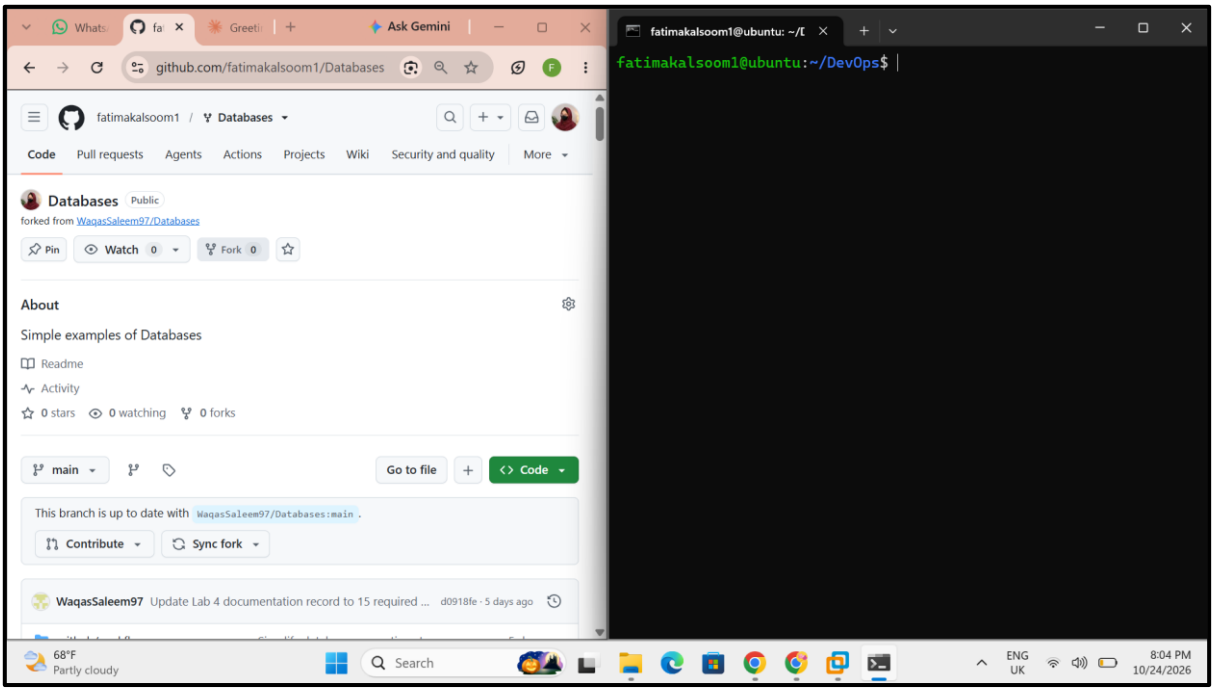


Figure 1: Split-screen view: my fork fatimakalsoom1/Databases on GitHub (forked from WaqasSaleem97/Databases) beside the Ubuntu terminal showing the fatimakalsoom1@ubuntu prompt.

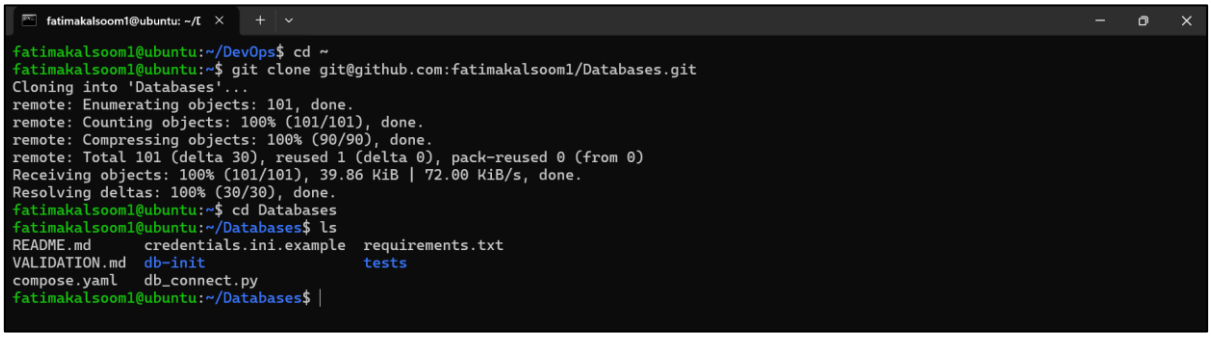


Figure 2: Cloning my fork over SSH, then ls inside ~/Databases showing compose.yaml, credentials.ini.example, db-init, db_connect.py, requirements.txt, README.md, VALIDATION.md and tests.

Task 2: Install Docker, Docker Compose and Python on Ubuntu

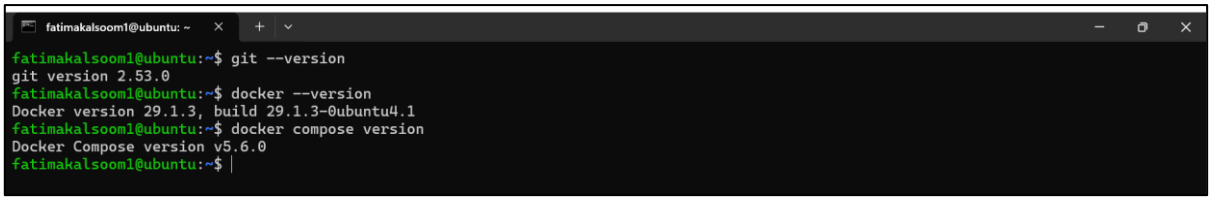


Figure 3: The three verification commands and their installed versions: Git 2.53.0, Docker 29.1.3 and Docker Compose v5.6.0.

Task 3: Start the Database Services



Figure 4: docker compose ps: mysql, mongo-dev, mongo-test and mongo-prod are all Up and (healthy).

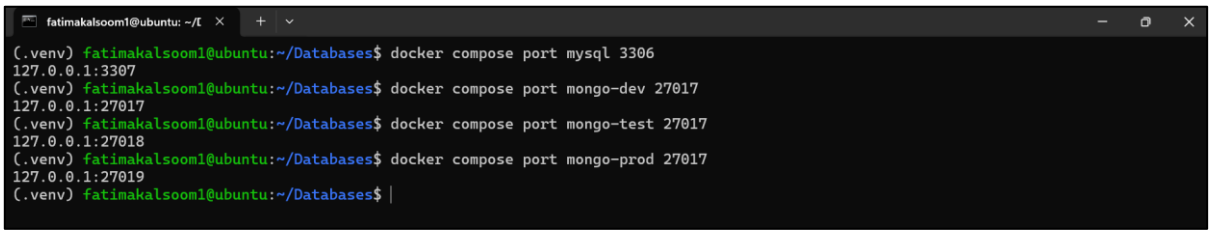


Figure 5: Host endpoints used by Python: MySQL 127.0.0.1:3307, MongoDB dev 27017, MongoDB test 27018 and MongoDB simulated prod 27019.

Database	Host endpoint	Database name	Environment
MySQL	127.0.0.1:3307	school_mysql	dev
MongoDB dev	127.0.0.1:27017	school_dev	dev
MongoDB test	127.0.0.1:27018	school_test	test
MongoDB simulated prod	127.0.0.1:27019	school_prod	prod

Task 4: MySQL with Values in the Code

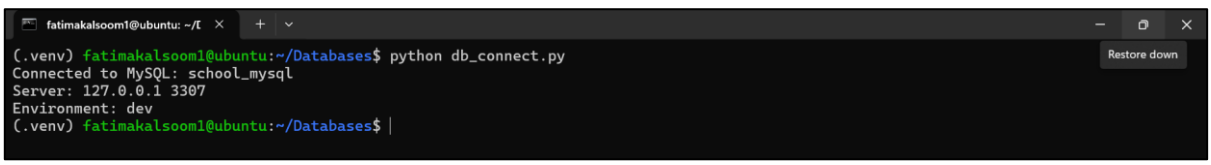


Figure 6: python db_connect.py with the MySQL code: Connected to MySQL: school_mysql, Server: 127.0.0.1 3307, Environment: dev.

Task 5: MongoDB with Values in the Code

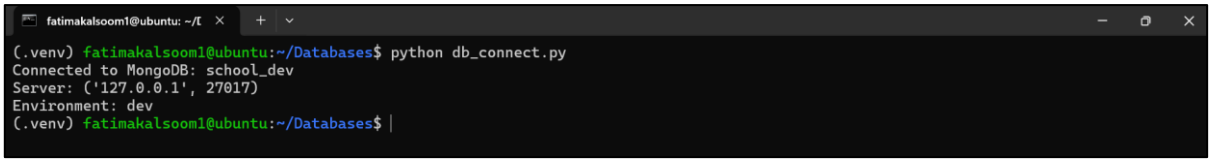


Figure 7: python db_connect.py with hardcoded MongoDB values: Connected to MongoDB: school_dev, Server: ('127.0.0.1', 27017), Environment: dev.

Task 6: MongoDB with os.getenv()

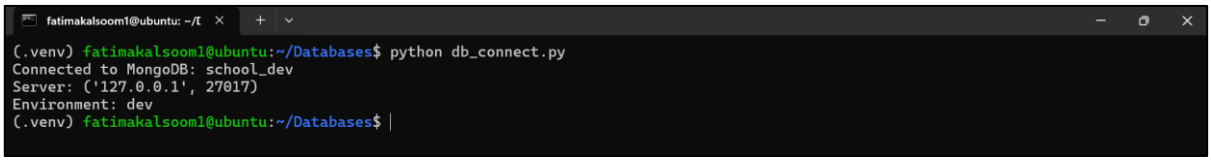


Figure 8: Environment-variable version with the dev profile: school_dev, port 27017, Environment: dev.

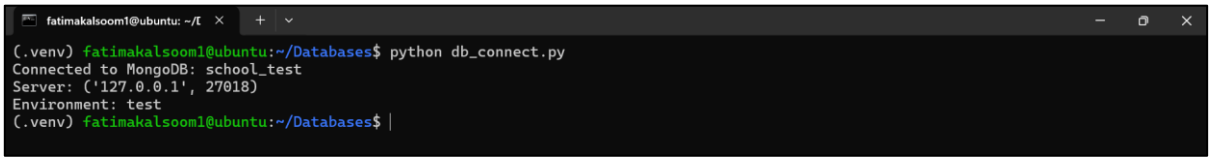


Figure 9: Same unchanged program with the test profile: school_test, port 27018, Environment: test.

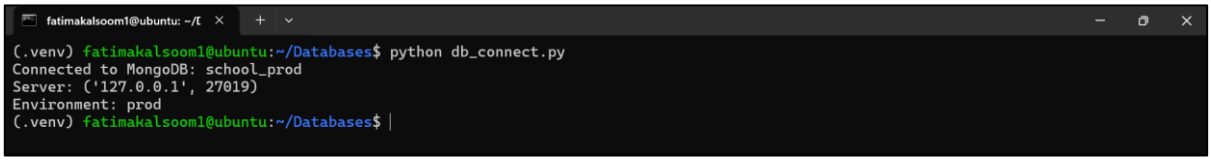


Figure 10: Same unchanged program with the simulated prod profile: school_prod, port 27019, Environment: prod.

Task 7: MongoDB with a Credentials File

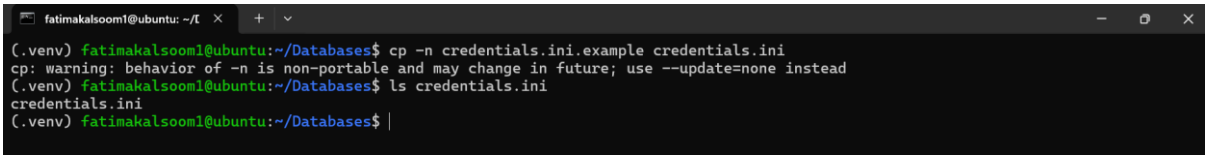


Figure 11: credentials.ini created from credentials.ini.example (the cp warning about -n is only a notice about future behaviour).

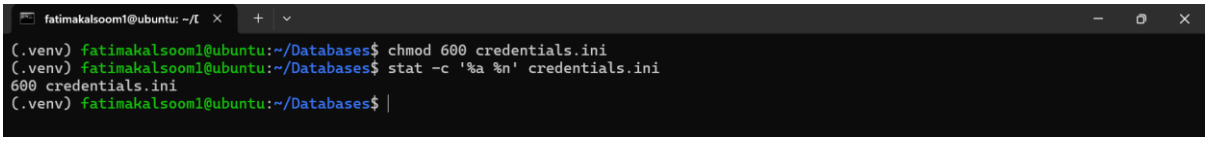


Figure 12: File mode set with chmod 600; stat reports 600 credentials.ini, so only the owner can read or write it.

I then replaced db_connect.py with the credentials-file version. It reads credentials.ini with configparser (interpolation=None so values are treated literally), selects a section using os.getenv("APP_ENV", "dev"), and reads the username, password, endpoint and database from that section. The MongoDB connection and output block are unchanged.

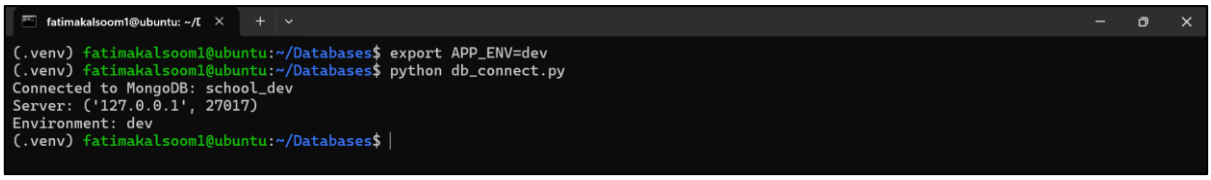


Figure 13: APP_ENV=dev: the dev section of the file is used, giving school_dev, port 27017, Environment: dev.

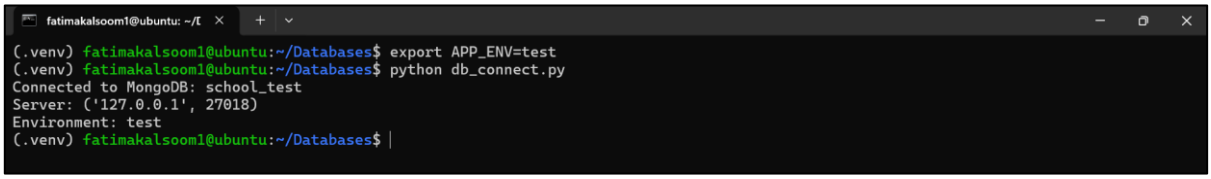


Figure 14: APP_ENV=test: the test section is used, giving school_test, port 27018, Environment: test.

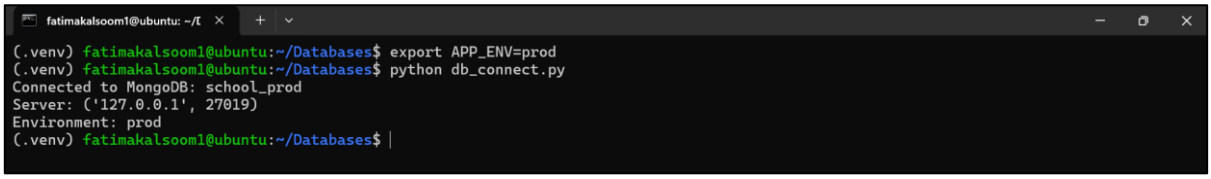


Figure 15: APP_ENV=prod: the prod section is used, giving school_prod, port 27019, Environment: prod.

Exam Evaluation Questions

Question 1: A Missing Environment Variable

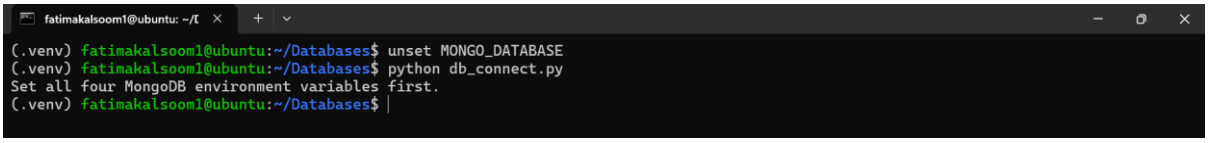


Figure 16: After unset MONGO_DATABASE the program stops with: Set all four MongoDB environment variables first.

Question 2: Which Source Does the File Version Read?

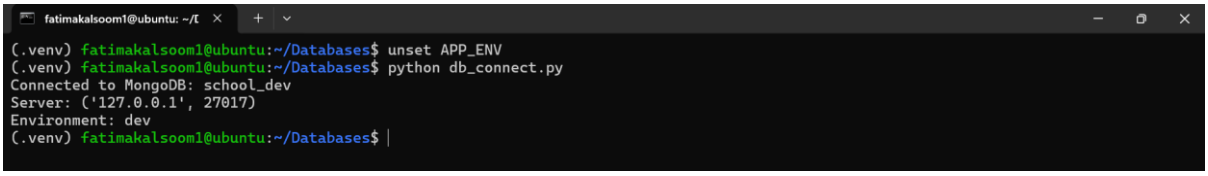


Figure 17: After unset APP_ENV the file version still connects to dev: school_dev, port 27017, Environment: dev.

Question 3: A Wrong Password in the File

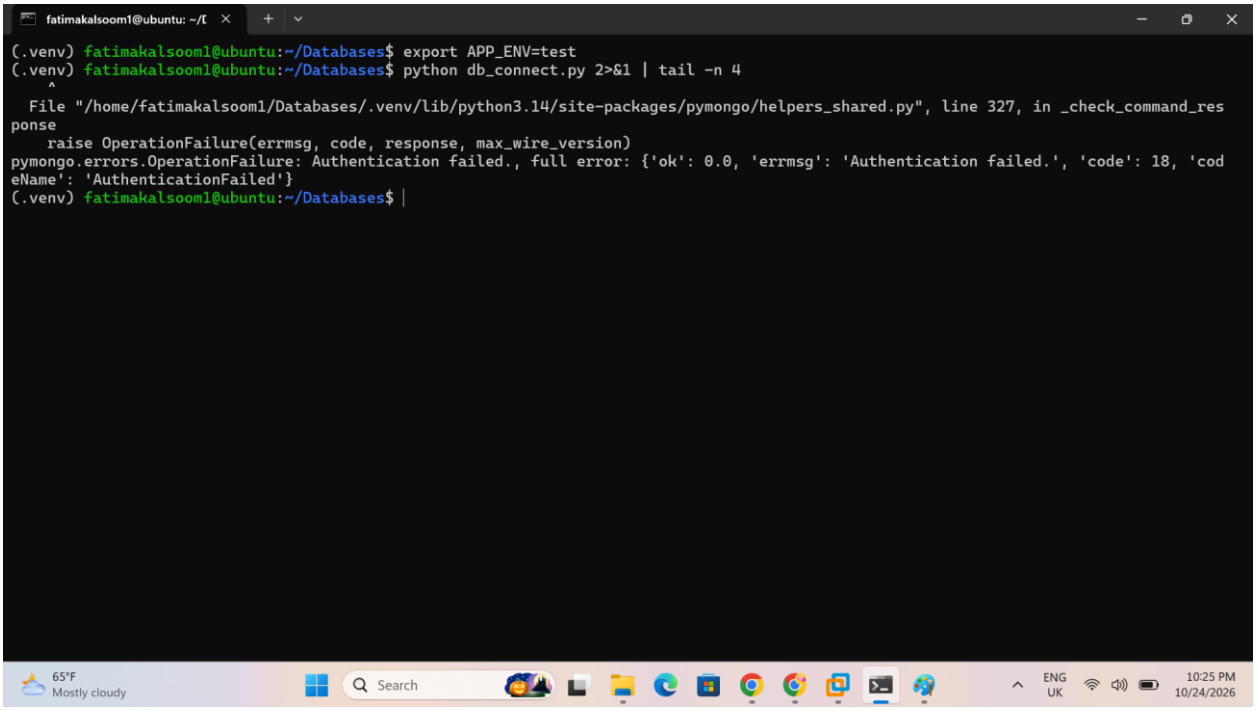


Figure 18: With a wrong test password the program fails inside pymongo while authenticating (the traceback is shown; the file itself is not displayed).
